

VisioCodicis: Disaster Routing Simulation

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Routing area: Bulacan, Philippines | A* vs. ABSRA (Arc-flag Bidirectional Search A*)

1. Baseline Synthetic Disaster Scenarios

Lvl	Start -> Goal	Algo	Cost (min)	Nodes	Time (ms)
L1 Minor	HQ_Malolos -> Hagonoy	A*	17.45	4	0.008
L1 Minor	HQ_Malolos -> Hagonoy	ABSRA	17.45	2	0.348
L1 Minor	HQ_Malolos -> San Miguel (Viola St)	A*	79.56	15	0.019
L1 Minor	HQ_Malolos -> San Miguel (Viola St)	ABSRA	79.56	3	0.347
L1 Minor	HQ_Malolos -> Bocaue (Crossing)	A*	21.33	8	0.012
L1 Minor	HQ_Malolos -> Bocaue (Crossing)	ABSRA	21.33	4	0.349
L1 Minor	HQ_Malolos -> Calumpit (Market)	A*	29.02	10	0.015
L1 Minor	HQ_Malolos -> Calumpit (Market)	ABSRA	29.02	3	0.349
L1 Minor	Plaridel -> Bocaue	A*	19.64	6	0.012
L1 Minor	Plaridel -> Bocaue	ABSRA	19.64	3	0.361
L2 Moderate	HQ_Malolos -> Hagonoy	A*	32.14	8	0.014
L2 Moderate	HQ_Malolos -> Hagonoy	ABSRA	32.14	3	0.355
L2 Moderate	HQ_Malolos -> San Miguel (Viola St)	A*	59.95	12	0.017
L2 Moderate	HQ_Malolos -> San Miguel (Viola St)	ABSRA	59.95	3	0.354
L2 Moderate	HQ_Malolos -> Bocaue (Crossing)	A*	29.36	10	0.016
L2 Moderate	HQ_Malolos -> Bocaue (Crossing)	ABSRA	29.36	4	0.355
L2 Moderate	HQ_Malolos -> Calumpit (Market)	A*	46.18	13	0.018
L2 Moderate	HQ_Malolos -> Calumpit (Market)	ABSRA	46.18	3	0.354
L2 Moderate	Plaridel -> Bocaue	A*	39.09	9	0.016
L2 Moderate	Plaridel -> Bocaue	ABSRA	39.09	4	0.356
L3 Severe	HQ_Malolos -> Hagonoy	A*	27.41	6	0.010
L3 Severe	HQ_Malolos -> Hagonoy	ABSRA	27.41	2	0.213
L3 Severe	HQ_Malolos -> San Miguel (Viola St)	A*	180.57	10	0.013
L3 Severe	HQ_Malolos -> San Miguel (Viola St)	ABSRA	180.57	3	0.216
L3 Severe	HQ_Malolos -> Bocaue (Crossing)	A*	INF	10	0.013
L3 Severe	HQ_Malolos -> Bocaue (Crossing)	ABSRA	INF	2	0.211

L3 Severe	HQ_Malolos -> Calumpit (Market)	A*	27.45	7	0.011
L3 Severe	HQ_Malolos -> Calumpit (Market)	ABSRA	27.45	3	0.213
L3 Severe	Plaridel -> Bocaue	A*	INF	10	0.013
L3 Severe	Plaridel -> Bocaue	ABSRA	INF	2	0.210

2. Flood Scenarios (Mamuyac 2025 threshold: 25 cm | NDRRMC Carina Jul 2024)

Lvl	Start -> Goal	MaxDepth(cm)	>25cm edges	Closed roads	A* (min)	ABSRA (min)	A* Nodes	ABSRA Nodes
L1	HQ_Malolos -> Hagonoy	20.0	0	0	13.60	13.60	4	2
L1	HQ_Malolos -> San Miguel (Viola St)	20.0	0	0	58.43	58.43	15	3
L1	HQ_Malolos -> Bocaue (Crossing)	20.0	0	0	22.49	22.49	10	4
L1	HQ_Malolos -> Calumpit (Market)	20.0	0	0	26.50	26.50	10	3
L1	Plaridel -> Bocaue	20.0	0	0	20.76	20.76	7	3
L2	HQ_Malolos -> Hagonoy	64.2	7	2	INF	INF	13	2
L2	HQ_Malolos -> San Miguel (Viola St)	64.2	7	2	69.25	69.25	13	3
L2	HQ_Malolos -> Bocaue (Crossing)	64.2	7	2	32.83	32.83	10	4
L2	HQ_Malolos -> Calumpit (Market)	64.2	7	2	INF	INF	13	2
L2	Plaridel -> Bocaue	64.2	7	2	44.34	44.34	10	4
L3	HQ_Malolos -> Hagonoy	105.8	15	12	INF	INF	1	1
L3	HQ_Malolos -> San Miguel (Viola St)	105.8	15	12	INF	INF	1	1
L3	HQ_Malolos -> Bocaue (Crossing)	105.8	15	12	INF	INF	1	1
L3	HQ_Malolos -> Calumpit (Market)	105.8	15	12	INF	INF	1	1
L3	Plaridel -> Bocaue	105.8	15	12	INF	INF	1	1

3. Heavy-Traffic Scenarios (BPR model | DPWH Region III / TomTom 2024)

Lvl	Start -> Goal	Max V/C	Edges >= 1.0	A* (min)	ABSRA (min)	A* Nodes	ABSRA Nodes	Blocked roads
L1	HQ_Malolos -> Hagonoy	1.00	9	13.45	13.45	4	2	0
L1	HQ_Malolos -> San Miguel (Viola St)	1.00	9	58.43	58.43	15	3	0
L1	HQ_Malolos -> Bocaue (Crossing)	1.00	9	21.67	21.67	10	4	0
L1	HQ_Malolos -> Calumpit (Market)	1.00	9	26.45	26.45	10	3	0
L1	Plaridel -> Bocaue	1.00	9	20.75	20.75	8	3	0
L2	HQ_Malolos -> Hagonoy	1.36	14	12.93	12.93	4	2	0
L2	HQ_Malolos -> San Miguel (Viola St)	1.36	14	71.04	71.04	15	3	0
L2	HQ_Malolos -> Bocaue (Crossing)	1.36	14	23.46	23.46	8	4	0

L2	HQ_Malolos -> Calumpit (Market)	1.36	14	31.97	31.97	10	3	0
L2	Plaridel -> Bocaue	1.36	14	24.00	24.00	6	3	0
L3	HQ_Malolos -> Hagonoy	1.69	15	20.59	20.59	4	2	0
L3	HQ_Malolos -> San Miguel (Viola St)	1.69	15	100.09	100.09	15	3	0
L3	HQ_Malolos -> Bocaue (Crossing)	1.69	15	39.74	39.74	11	4	0
L3	HQ_Malolos -> Calumpit (Market)	1.69	15	41.74	41.74	10	3	0
L3	Plaridel -> Bocaue	1.69	15	32.98	32.98	8	3	0

4. Road-Block Scenarios (DPWH Region III | Bulacan DEO post-typhoon bulletins)

Lvl	Start -> Goal	Max S	Edges >= 0.6	Closed roads	A* (min)	ABSRA (min)	A* Nodes	ABSRA Nodes
L1	HQ_Malolos -> Hagonoy	0.24	0	0	15.54	15.54	4	2
L1	HQ_Malolos -> San Miguel (Viola St)	0.24	0	0	63.34	63.34	15	3
L1	HQ_Malolos -> Bocaue (Crossing)	0.24	0	0	21.33	21.33	8	4
L1	HQ_Malolos -> Calumpit (Market)	0.24	0	0	36.08	36.08	11	3
L1	Plaridel -> Bocaue	0.24	0	0	19.64	19.64	6	3
L2	HQ_Malolos -> Hagonoy	0.54	0	0	12.30	12.30	4	2
L2	HQ_Malolos -> San Miguel (Viola St)	0.54	0	0	104.87	104.87	15	3
L2	HQ_Malolos -> Bocaue (Crossing)	0.54	0	0	38.71	38.71	11	4
L2	HQ_Malolos -> Calumpit (Market)	0.54	0	0	47.90	47.90	11	3
L2	Plaridel -> Bocaue	0.54	0	0	26.91	26.91	6	3
L3	HQ_Malolos -> Hagonoy	0.70	9	9	INF	INF	5	2
L3	HQ_Malolos -> San Miguel (Viola St)	0.70	9	9	INF	INF	5	2
L3	HQ_Malolos -> Bocaue (Crossing)	0.70	9	9	INF	INF	5	1
L3	HQ_Malolos -> Calumpit (Market)	0.70	9	9	INF	INF	5	2
L3	Plaridel -> Bocaue	0.70	9	9	INF	INF	5	1

5. Algorithm Comparison Summary (averaged per scenario type & level)

Scenario	Level	Algorithm	Avg Cost (min)	Avg Nodes	Avg Time (ms)	Reachable
Generic	L1	A*	33.40	8.6	0.013	5/5
Generic	L1	ABSRA	33.40	3.0	0.351	5/5
Generic	L2	A*	41.34	10.4	0.016	5/5
Generic	L2	ABSRA	41.34	3.4	0.355	5/5
Generic	L3	A*	78.48	7.7	0.011	3/5
Generic	L3	ABSRA	78.48	2.7	0.214	3/5
Flood	L1	A*	28.35	9.2	0.014	5/5
Flood	L1	ABSRA	28.35	3.0	0.352	5/5
Flood	L2	A*	48.81	11.0	0.016	3/5
Flood	L2	ABSRA	48.81	3.7	0.297	3/5
Flood	L3	A*	-	-	-	0/5
Flood	L3	ABSRA	-	-	-	0/5
Traffic	L1	A*	28.15	9.4	0.021	5/5
Traffic	L1	ABSRA	28.15	3.0	0.370	5/5
Traffic	L2	A*	32.68	8.6	0.013	5/5
Traffic	L2	ABSRA	32.68	3.0	0.352	5/5
Traffic	L3	A*	47.03	9.6	0.015	5/5
Traffic	L3	ABSRA	47.03	3.0	0.360	5/5
Road-Block	L1	A*	31.19	8.8	0.013	5/5
Road-Block	L1	ABSRA	31.19	3.0	0.349	5/5
Road-Block	L2	A*	46.14	9.4	0.014	5/5
Road-Block	L2	ABSRA	46.14	3.0	0.352	5/5
Road-Block	L3	A*	-	-	-	0/5
Road-Block	L3	ABSRA	-	-	-	0/5